

Shuaijun Wang

SENIOR ROBOTICS ALGORITHM ENGINEER · DEXTEROUS MANIPULATION TECHNICAL LEAD

Beijing, China

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Summary

Senior robotics algorithm engineer and Ph.D. in robotic learning, currently serving as Dexterous Manipulation Technical Lead at Xiaomi Robotics. Specializes in dexterous grasping and manipulation, tactile perception and feedback, reinforcement and imitation learning, whole-body autonomous manipulation, and embodied AI. Experienced in training, evaluating, and iterating multi-task VLA/WAM foundation models on large-scale compute clusters, with end-to-end leadership spanning simulation, data collection, foundation-model and policy learning, sim-to-real transfer, and deployment on humanoid platforms.

Education

Harbin Institute of Technology / The University of Edinburgh (CSC)

PH.D. IN ROBOTIC LEARNING

Harbin / Edinburgh

Sep. 2018 – May. 2024

- Joint Ph.D. program funded by China Scholarship Council (CSC).
- Supervised by Prof. Lining Sun and Prof. Zhibin (Alex) Li.
- Researched adaptive learning and control for robotic grasping and manipulation.
- Visiting researcher at the Edinburgh Centre for Robotics from Sep. 2019 to Sep. 2021; published in *IEEE/ASME Transactions on Mechatronics*.
- Collaborated remotely with Prof. Qiang Li at Bielefeld University from Mar. to Dec. 2022 on calibration-free visual servoing; published in *International Journal of Humanoid Robotics*.

Harbin Institute of Technology (Shenzhen)

M.SC. IN MECHATRONIC ENGINEERING

Shenzhen, China

Sep. 2013 – Sep. 2015

- Supervised by Prof. Xin Wang; focused on modular robotic arm design, kinematics, and manipulator control.

Northeast Forestry University (Project 211)

B.ENG. IN INTELLIGENT EQUIPMENT ENGINEERING (TOP 5%)

Harbin, China

Sep. 2009 – Sep. 2013

- Awarded Outstanding Graduate.

Work Experience

Xiaomi Robotics

SENIOR ALGORITHM ENGINEER / DEXTEROUS MANIPULATION TECHNICAL LEAD

Beijing, China

Jul. 2024 – Present

- Selected for Xiaomi Future Star, the company's top-tier talent program.
- Built the dexterous manipulation team from 0 to 1 and serve as technical lead for humanoid grasping and manipulation algorithms.
- Lead research and deployment across tactile grasping, in-hand and bimanual manipulation, imitation/reinforcement learning, and dexterous-manipulation foundation models.

Tencent Robotics X Lab

RESEARCH INTERN

Shenzhen, China

Dec. 2022 – Jun. 2023

- Contributed to dexterous in-hand manipulation, tactile perception, and visuo-tactile multimodal manipulation algorithms.
- Trained policies in simulation and completed sim-to-real transfer and real-robot validation.

Shenzhen Institute of Aerospace Technology Innovation

ALGORITHM ENGINEER (ROBOTICS)

Shenzhen, China

Jul. 2016 – Aug. 2018

- Developed QP-based kinematics, trajectory planning, and force-sensor-based teaching by demonstration.
- Built model-based grasping pipelines with Mask R-CNN for segmentation and 4PCS for pose estimation.
- Integrated RealSense cameras with UR arms for hand-eye calibration, perception, and robot control.
- Video: YouTube link.

AVIC 613 Research Institute

ENGINEER

Luoyang, China

Sep. 2015 – Apr. 2016

- Mechanical design and fluid simulation for electro-optical radar systems.

Project Experience

Embodied Foundation Models for Dexterous Manipulation

TECHNICAL LEAD

Beijing, China

Mar. 2026 – Present

- Built an Xsens + MANUS teleoperation system for full-body and hand motion capture, including motion retargeting and coordinated control of RealMan dual arms and Sharpa dexterous hands.
- Established an end-to-end pipeline covering teleoperation, robot execution, data logging, and quality validation, enabling efficient collection of natural, human-like bimanual dexterous-manipulation data.
- Developed multi-task VLA/WAM embodied foundation models that jointly model vision, language, and arm-hand actions for cross-task training and policy reuse.
- Built a simulation-based dexterous-manipulation benchmark with standardized tasks and evaluation protocols; the developed models outperformed the $\pi_{0.5}$ baseline overall.

Visuo-Tactile Dexterous Manipulation Algorithm Development

TECHNICAL LEAD

Beijing, China

Jul. 2024 – Present

- Led end-to-end R&D for tactile grasping, in-hand manipulation, bimanual coordination, and visuo-tactile learning, from data collection and policy training to sim-to-real transfer and deployment.
- Developed TacRefineNet, a tactile-only closed-loop grasp-refinement policy for precise transitions between arbitrary in-hand object poses, reducing reliance on external vision.
- TacRefineNet was featured in Xiaomi Group's FY2025 annual results materials and established a reusable tactile perception-and-control pipeline.
- Developed ViTacPhys to infer object mass, stiffness, and friction from human visual-tactile demonstrations and adapt dexterous grasping policies to the estimated physical properties.
- Developed RGB-only dynamic bimanual handover and advanced high-speed bimanual throwing, continuous catching, and three-ball juggling on humanoid robots.
- Related results were submitted to IEEE RA-L and IEEE T-ASE and are currently under review.

Reinforcement Learning for Tactile In-Hand Manipulation

LEAD RESEARCHER

Shenzhen, China

Dec. 2022 – Jun. 2023

- Built a contact-rich in-hand manipulation environment in MuJoCo and trained dexterous-hand policies with PPO, including observation, action, and reward design.
- Integrated real-time fingertip tactile signals into policy observations and closed-loop control, enabling responsive finger adjustment and more stable manipulation.
- Used CMA-ES to optimize PD gains and reduce the sim-to-real gap, transferring the policy to a real robot for tool-based stirring; the result was publicly demonstrated in 2023.

Visual Servoing for Adaptive Arm-Hand Grasping

LEAD RESEARCHER

Shenzhen, China

Mar. 2022 – Dec. 2022

- Developed calibration-free grasping using only low-cost RGB cameras, eliminating camera intrinsic/extrinsic and hand-eye calibration requirements.
- Learned a closed-loop mapping from image observations to robot motion commands, continuously correcting end-effector pose errors from visual feedback.
- Built an end-to-end visual-servoing framework for real-time 3D grasping across diverse objects; published in *International Journal of Humanoid Robotics*.

Reinforcement Learning for Contact-Aware Object Manipulation

LEAD RESEARCHER

Edinburgh / Shenzhen

Dec. 2021 – Jul. 2022

- Formulated contact-aware reinforcement learning for robotic reaching and object pushing by explicitly incorporating contact states into policy observations and control.
- Trained continuous-control policies with Soft Actor-Critic to adapt motion direction and applied force based on contact feedback.
- Validated generalization and training efficiency across target positions and object conditions; published in *Frontiers in Neuro-robotics* (SCI Q2).

Adaptive Dexterous Hand Grasp Control via Learning from Human Demonstration

LEAD RESEARCHER

Edinburgh, UK

Sep. 2019 – Dec. 2020

- Built a human teleoperation and data-collection pipeline and gathered grasp demonstrations across objects with varied sizes, geometries, and stiffness.
- Trained supervised adaptive grasping models to learn the mapping among object properties, hand postures, and grasping strategies.
- Demonstrated robust cross-object generalization on unseen objects; published in *IEEE/ASME Transactions on Mechatronics* (SCI Q1).

Optimization-Based Manipulator Control and Force-Guided Direct Teaching

CORE DEVELOPER

Shenzhen, China

Jul. 2016 – Aug. 2018

- Built a generic quadratic-programming control framework that formulated trajectory tracking, joint limits, velocity bounds, and obstacle avoidance as an online optimization problem.
- Implemented joint-space and Cartesian-space control for stable trajectory planning and motion control under multiple constraints.
- Developed force-sensor-based direct teaching with tool-gravity compensation for intuitive drag teaching and trajectory recording.
- Published at IEEE ICARM 2017 and IEEE CYBER 2019 (both EI indexed).

Rapidly Reconfigurable Modular Space Manipulator

CORE DEVELOPER

Shenzhen, China

Sep. 2013 – Sep. 2015

- Designed the system architecture, modular joints, and mechanical interfaces for rapid connection and separation in on-orbit manipulation.
- Developed kinematic models and workspace analyses for different module combinations, enabling task-oriented configuration and rapid reconfiguration.
- Published in *Applied Mechanics & Materials* (EI indexed) and granted Chinese invention patent CN104647396A.

Publications & Patents

1. Anonymous authors and **Wang, Shuaijun** (Last/corresponding author). “ViTacPhys: Physical Property-Aware Grasping from Human Visual-Tactile Demonstrations.” IEEE T-ASE, under review, 2026.
2. **Wang, Shuaijun**, et al. “Dexterous-Hand Fine Grasping via Proprioceptive-Visual-Tactile Multimodal Fusion.” Chinese and U.S. patent applications pending, 2026.
3. **Wang, Shuaijun**, et al. “Dexterous-Hand Fine Grasping via Multi-Finger Tactile Fusion.” Chinese patent application pending, 2026.

4. **Wang, Shuaijun** (Corresponding author), Haoran Zhou, Diyun Xiang, and Yangwei You. “*TacRefineNet: Tactile-Only Grasp Refinement Between Arbitrary In-Hand Object Poses.*” arXiv:2509.25746, 2025. (IEEE RA-L, under review)
5. Zhou, Haoran, Yangwei You, and **Wang, Shuaijun** (Corresponding author). “*DyDexHandover: Human-like Bimanual Dynamic Dexterous Handover using RGB-only Perception.*” arXiv:2509.17350, 2025.
6. **Wang, Shuaijun**, Lining Sun, Mantian Li, et al. “*Learning an Image-Based Visual Servoing Controller for Object Grasping.*” **International Journal of Humanoid Robotics (IJHR)**, 2024. (SCI)
7. **Wang, Shuaijun**, Lining Sun, Fusheng Zha, Wei Guo, Pengfei Wang. “*Learning Adaptive Reaching and Pushing Skills Using Contact Information.*” **Frontiers in Neurorobotics**, 2023. (SCI Q2)
8. **Wang, Shuaijun**, Wenbin Hu, Lining Sun, Xin Wang, Zhibin Li. “*Learning Adaptive Grasping From Human Demonstrations.*” **IEEE/ASME Transactions on Mechatronics (TMECH)**, 2022. (SCI Q1)
9. Wu, Jingyuan, Xin Wang, **Wang, Shuaijun**, et al. “*An Optimization-Based Direct Teaching Method for Manipulator with Tool’s Gravity Compensation.*” In **IEEE CYBER**, 2019. (EI Indexed)
10. **Wang, Shuaijun**, Lining Sun, Xin Wang, et al. “*A Generic Control Method of Manipulator Based on Optimization.*” In **Proc. of the 2nd IEEE ICARM**, 2017. (EI Indexed)
11. **Wang, Shuaijun**, Bo Zhang, Xin Wang, et al. “*Design and Research of a Novel Fast Connecting and Separating Modular Space Manipulator.*” **Applied Mechanics & Materials**, 2015. (EI Indexed)
12. Xueqian Wang, **Wang, Shuaijun**, Bo Zhang, Bin Liang, Yu Zou. “*Modular Space Robotic Arm Joints and Robotic Arms That Can Be Used for Rapid Reconfiguration.*” **Chinese Patent**, CN104647396A, 2015.

Academic Services

Conferences: ICRA (2024), IROS (2021, 2022, 2026), ICARM (2019)

REVIEWER

Journals: IEEE RAL; IEEE/ASME TMECH; International Journal of Humanoid Robotics; Industrial Robot; Robotic Intelligence and Automation

REVIEWER

Invited Talks

Beijing Academy of Artificial Intelligence · Embodied Intelligence Research Center

ADAPTIVE ROBOTIC GRASPING AND MANIPULATION VIA EMBODIED INTELLIGENCE

Beijing, China

Sep. 2023

- Invited by Prof. He Wang.

Shenzhen Technology University · School of Big Data and Internet

EMBODIED INTELLIGENCE METHODS FOR GENERAL-PURPOSE ROBOTIC GRASPING AND MANIPULATION

Shenzhen, China

Dec. 2023

- Invited by Prof. Qiang Li; received Assistant Professor offer following the talk.

Honors & Awards

2024	Xiaomi Future Star , Xiaomi Corporation top-tier talent program	Beijing, China
2019–21	CSC Government Scholarship for Overseas Study , China Scholarship Council (CSC)	China
2013	Outstanding Graduate , Northeast Forestry University	Harbin, China
2011	National Encouragement Scholarship , Ministry of Education, P.R. China	China